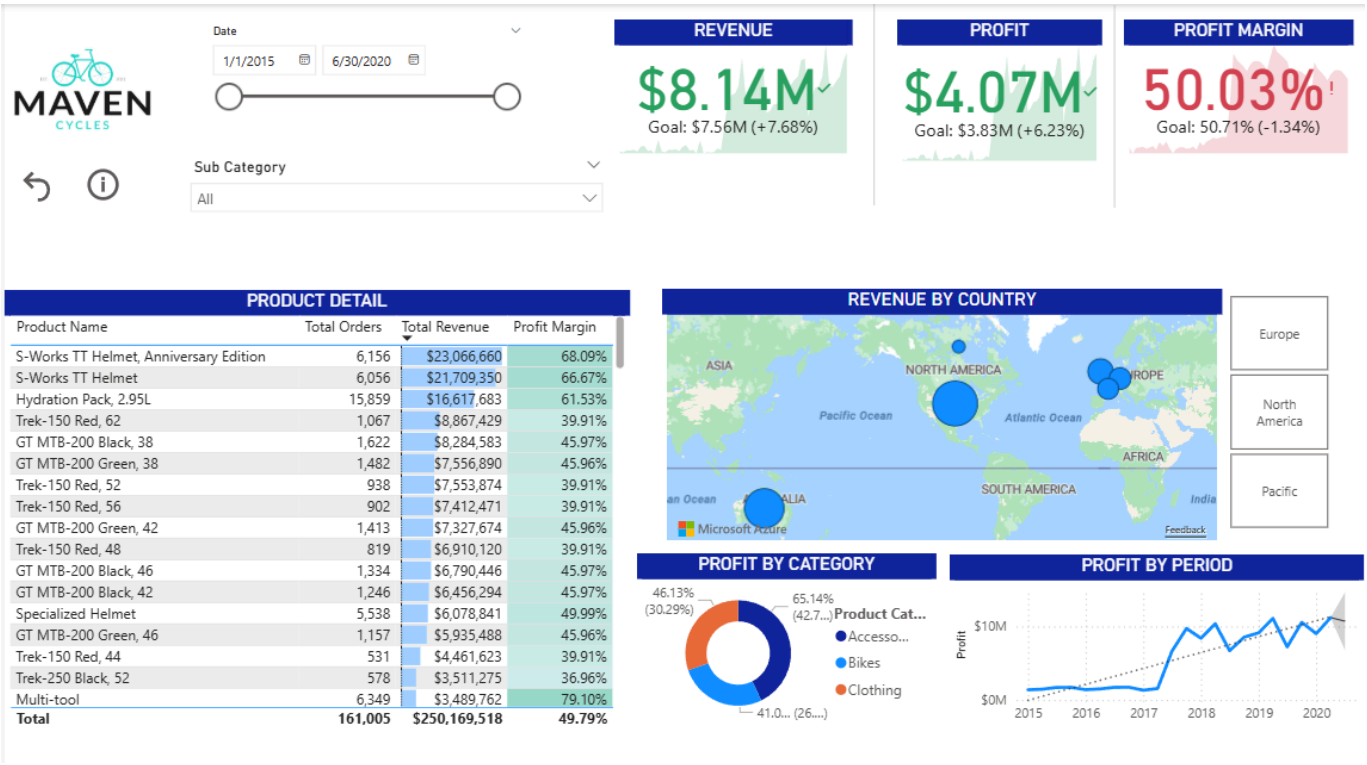
# Maven Cycles Sales Performance Analysis

**Dataset:** Maven Analytics Bicycle Retail Chain




| Product Name | Total Orders | Total Revenue | Profit Margin |
|---|---|---|---|
| S-Works TT Helmet, Anniversary Edition | 6,156 | $23,066,660 | 68.09% |
| S-Works TT Helmet | 6,056 | $21,709,350 | 66.67% |
| Hydration Pack, 2.95L | 15,859 | $16,617,683 | 61.53% |
| Trek-150 Red, 62 | 1,067 | $8,867,429 | 39.91% |
| GT MTB-200 Black, 38 | 1,622 | $8,284,583 | 45.97% |
| GT MTB-200 Green, 38 | 1,482 | $7,556,890 | 45.96% |
| Trek-150 Red, 52 | 938 | $7,553,874 | 39.91% |
| Trek-150 Red, 56 | 902 | $7,412,471 | 39.91% |
| GT MTB-200 Green, 42 | 1,413 | $7,327,674 | 45.96% |
| Trek-150 Red, 48 | 819 | $6,910,120 | 39.91% |
| GT MTB-200 Black, 46 | 1,334 | $6,790,446 | 45.97% |
| GT MTB-200 Black, 42 | 1,246 | $6,456,294 | 45.97% |
| Specialized Helmet | 5,538 | $6,078,841 | 49.99% |
| GT MTB-200 Green, 46 | 1,157 | $5,935,488 | 45.96% |
| Trek-150 Red, 44 | 531 | $4,461,623 | 39.91% |
| Trek-250 Black, 52 | 578 | $3,511,275 | 36.96% |
| Multi-tool | 6,349 | $3,489,762 | 79.10% |
| **Total** | **161,005** | **$250,169,518** | **49.79%** |

## Project Summary

Built a Power BI sales dashboard to analyze revenue, profitability, and regional performance for a multi region bicycle retail chain. The solution covers data cleaning, dimensional modeling, KPI development, secure deployment, and interactive analysis for business stakeholders. Additional screenshots below the report.

## Data Preparation

- Cleaned and standardized data using Power Query
- Consolidated multi year sales data
- Created additional columns including price range and aggregated product quantities
- Built a robust Calendar table with year, month, quarter, and time intelligence columns

## Data Modeling

- Designed a star schema with Sales as the fact table
- Connected Products, Categories, Subcategories, Calendar, Regions, Gender, and Age Group tables via one-to-many relationships
- Implemented an inactive date relationship to support advanced time analysis
- Ensured single direction filter flow for accurate category filtering

## Dashboard & Analysis

- KPI cards for Revenue, Profit, and Profit Margin with conditional formatting
- Interactive slicers for date and product subcategory
- Product-level table showing revenue contribution and margin
- Visual analysis of profit by category, region, and time period
- Drill-through pages for country-level product performance
- Bookmarks for pre-filtered executive views

## Deployment & Security

- Published to Power BI Service
- Configured scheduled refresh via data gateway
- Implemented static and dynamic Row Level Security (RLS) by region
- Distributed insights through a published Power BI App

# Quantifiable Insights

1. **Product Concentration Risk**

    - The top 20% of products generated ~69% of total revenue, indicating heavy reliance on a small subset of SKUs.

    - Recommendation: protect inventory availability for top performers while diversifying mid-tier product sales.

2. **Profitability vs. Volume Tradeoff**

    - Certain subcategories produced 30 - 40% higher profit margins  despite lower unit sales compared to high volume products.

    - Recommendation: prioritize marketing and promotions for high margin categories rather than purely high volume items.

3. **Regional Performance Gap**

    - The region with the lowest profit margin is ~10% lower than the highest performing region.

    - Recommendation: analyze pricing strategy, product mix, and regional demand drivers to close the profitability gap through the promotion of higher profit margin SKUs (accessories).

# Skills Demonstrated

- Power Query (data cleaning & transformation)
- Data modeling
- KPI design and profitability analysis
- Time based analysis and drill through
- Power BI Service deployment and governance
- Row Level Security implementation

# Screenshots

## Conditional column with formula

```
= Table.AddColumn(#"Renamed Columns", "Price Range", each if [Unit Price] <= 100 then "Low" else if [Unit Price] < 1000 then "Mid" else if [Unit Price] >=
    1000 then "High" else "Unclassified")
```

| | Product Key | Product Subcategory Key | Product New | Unit Cost | Unit Price | Price Range |
|---|---|---|---|---|---|---|
| 1 | 201 | 8 | Kit, Patch | 0.97 | 1.99 | Low |
| 2 | 202 | 8 | Tube, Tire Road | 1.49 | 4.99 | Low |
| 3 | 203 | 8 | Tube, Touring | 2.25 | 5.49 | Low |
| 4 | 204 | 8 | Tube, Mountain | 2.25 | 5.49 | Low |
| 5 | 205 | 8 | Mountain Tire, High Load | 12.99 | 39.99 | Low |
| 6 | 206 | 8 | Road Tire, All | 7.99 | 24.99 | Low |
| 7 | 207 | 8 | Road Tire, High Load | 12.99 | 33.49 | Low |
| 8 | 208 | 8 | Touring Tire | 10.99 | 29.99 | Low |

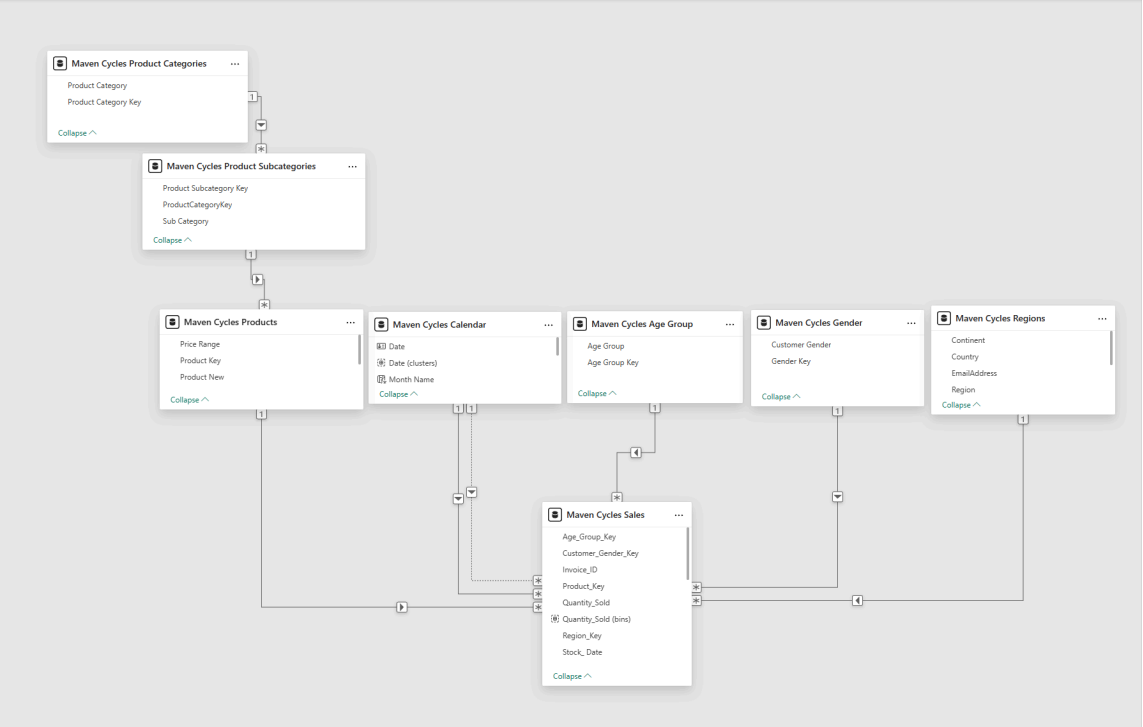
## Calendar Table

Name: Maven Cycles Cale...

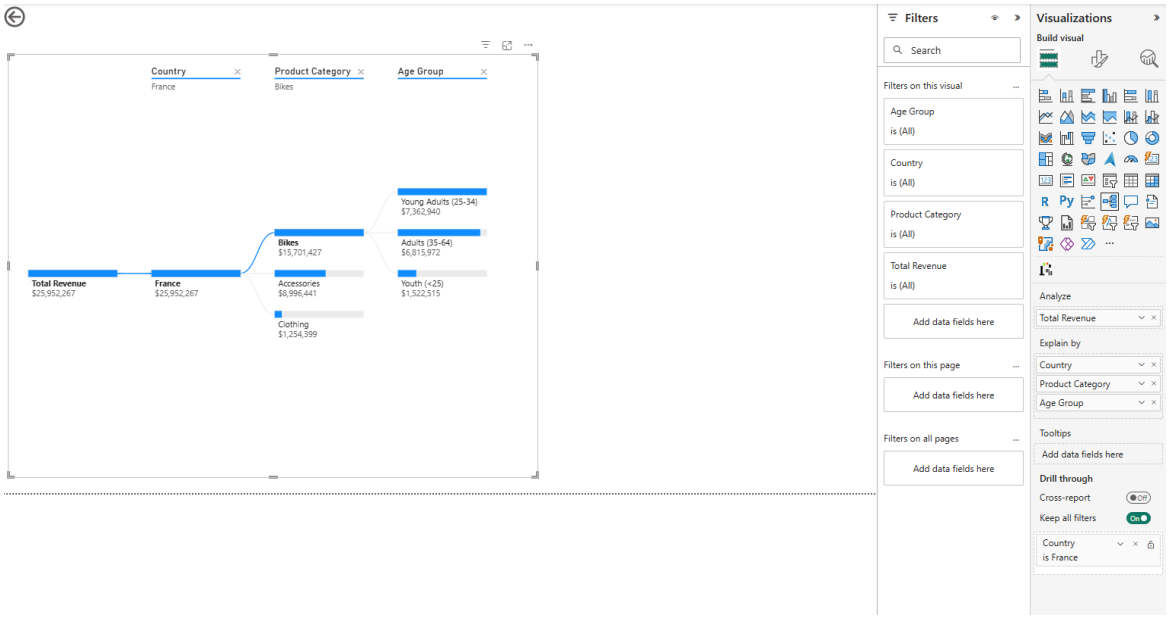
Structure | Manage relationships (Relationships) | New measure, Quick measure, New column, New table (Calculations) | Mark as date table (Calendars)

| Date | Year | Sart of Quarter | Start of Month | Month Number | Month Name | Date (clusters) |
|---|---|---|---|---|---|---|
| Thursday, January 1, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Friday, January 2, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Saturday, January 3, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Sunday, January 4, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Monday, January 5, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Tuesday, January 6, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Wednesday, January 7, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Thursday, January 8, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Friday, January 9, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Saturday, January 10, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Sunday, January 11, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |
| Monday, January 12, 2015 | 2015 | 1/1/2015 12:00:00 AM | January 2015 | 1 | Jan | Cluster1 |

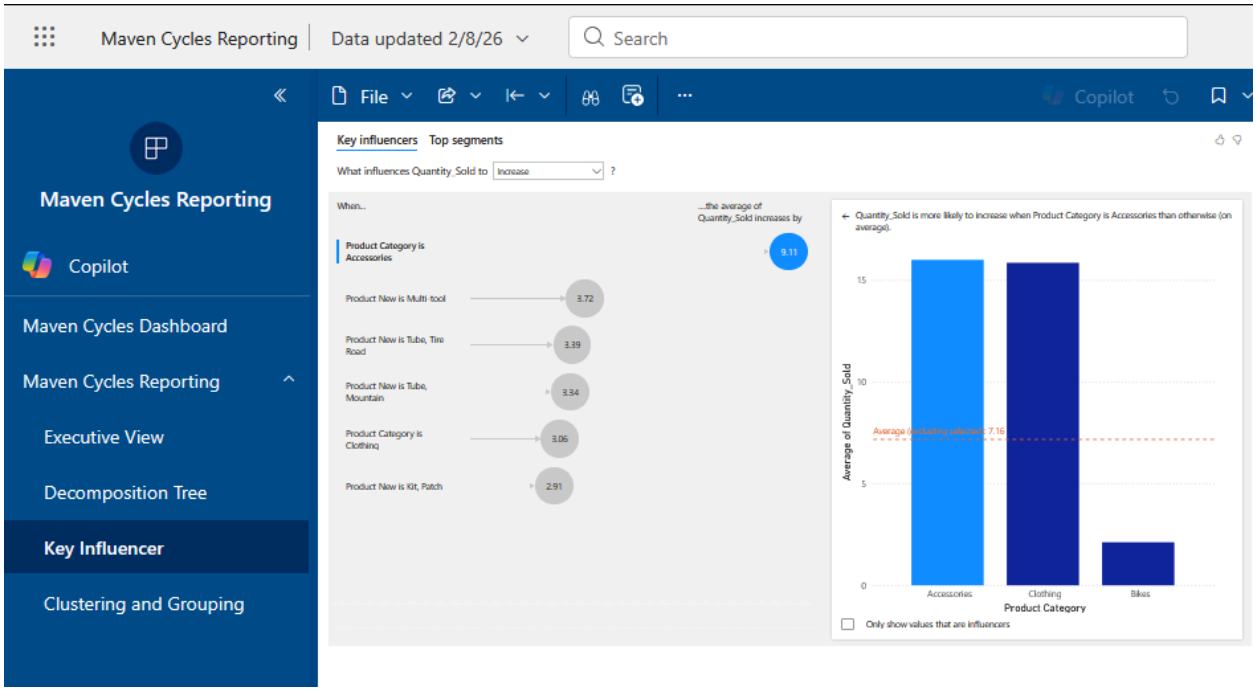
## Data Model (star schema)

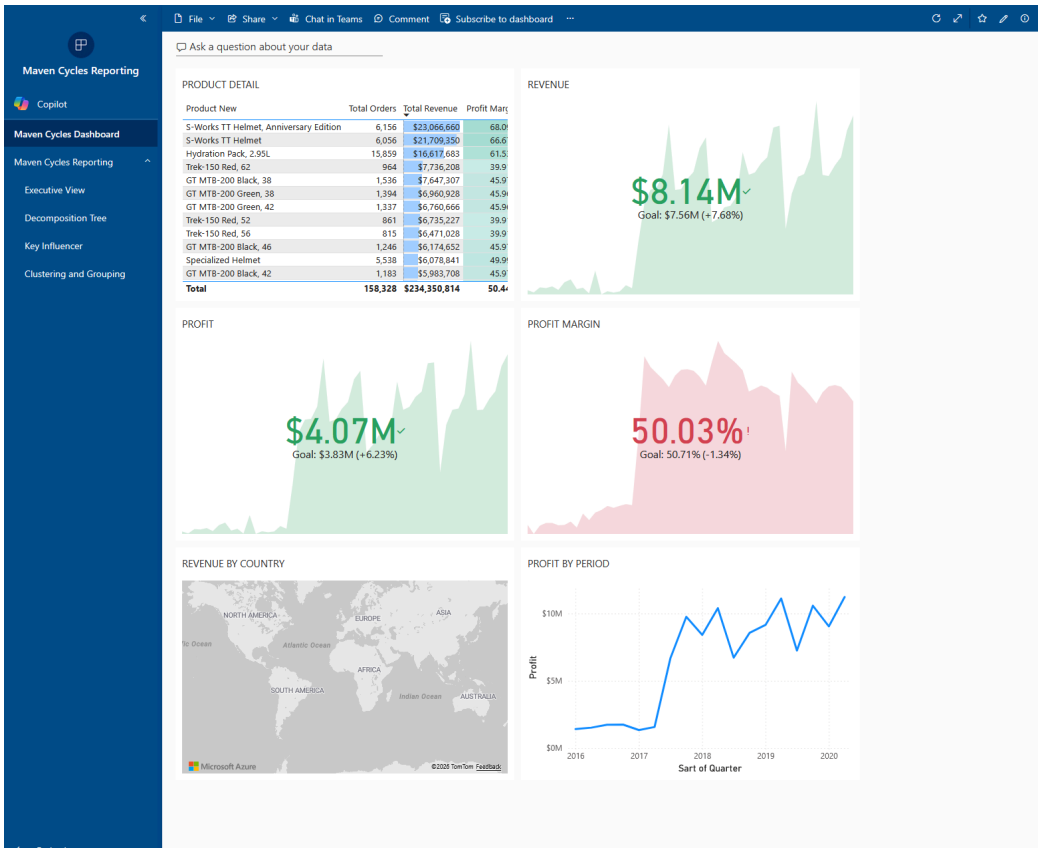



## Drill through filtering using Decomposition Tree

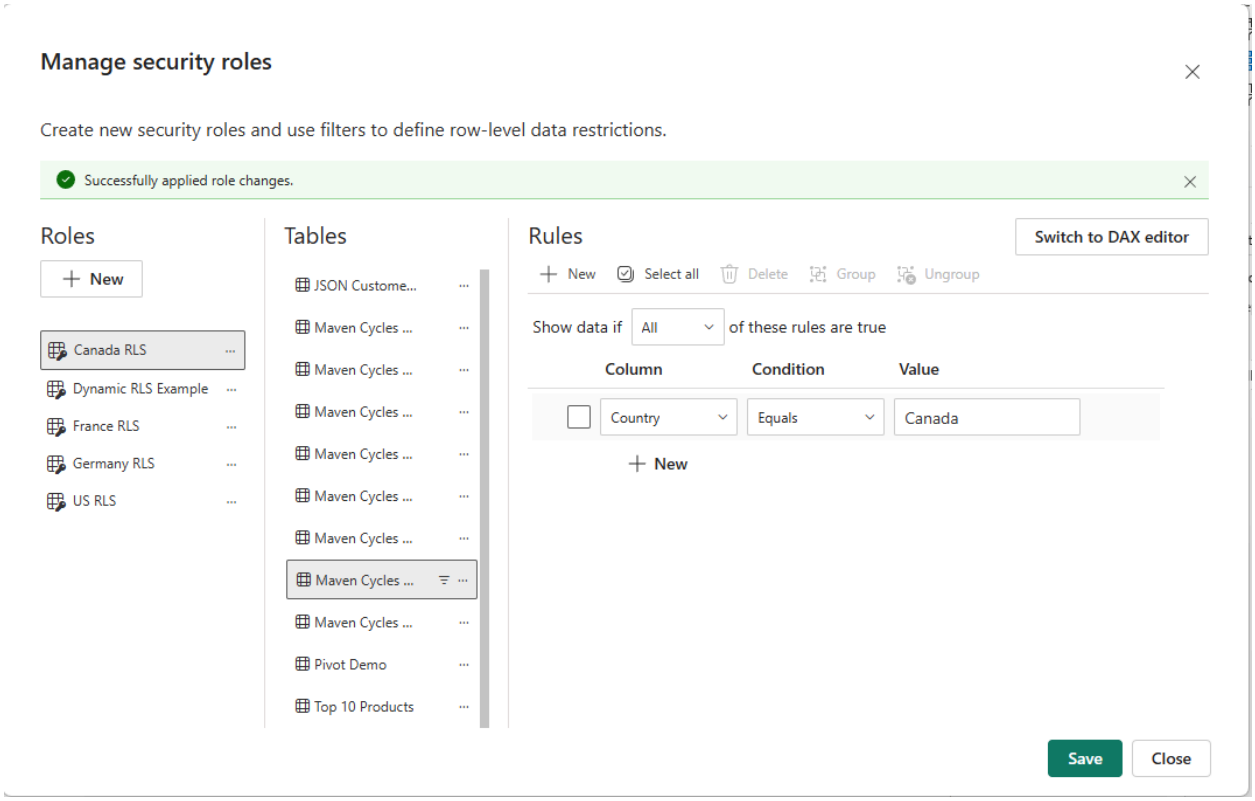


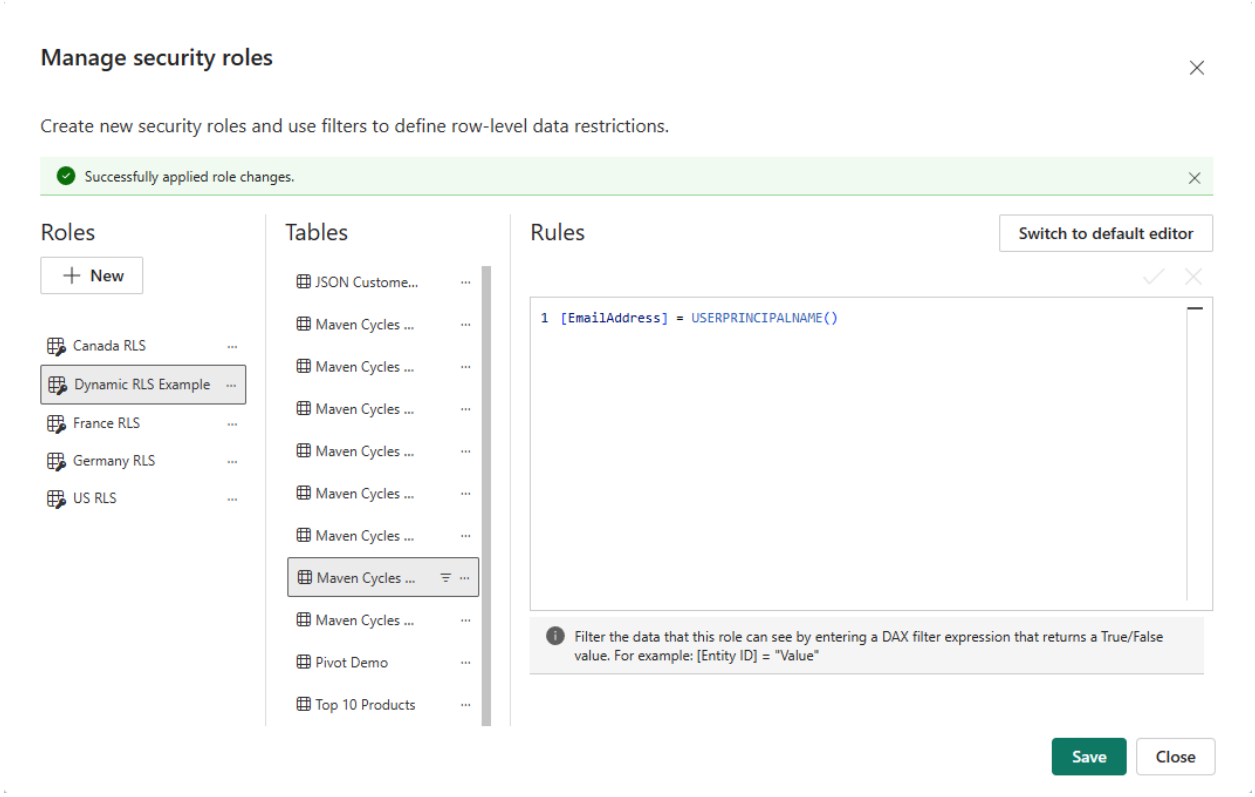
## Configured app



## Static RLS




## Dynamic RLS